

Step 1: Build Docker Image (docker build)

```
bash - docker build -t flask-sample-app:1.0 .
```

```
admin@dev-machine:~/lab2/q1_flask_app$ docker build -t flask-sample-app:1.0 .
[+] Building 4.8s (10/10) FINISHED
=> [internal] load build definition from Dockerfile                                0.0s
=> => transferring dockerfile: 382B
=> [internal] load metadata for docker.io/library/python:3.11-slim                1.2s
=> [internal] load .dockerignore                                                  0.0s
=> [1/5] FROM docker.io/library/python:3.11-slim@sha256:d8c520...                1.4s
=> => resolve docker.io/library/python:3.11-slim@sha256:d8c520...
=> [2/5] WORKDIR /app                                                            0.1s
=> [3/5] COPY requirements.txt .                                                  0.1s
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt                      1.8s
=> => Collecting flask==3.0.3
=> => Successfully installed Werkzeug-3.0.3 blinker-1.8.2 click-8.1.7 flask-3.0.3
=> [5/5] COPY app.py .                                                          0.1s
=> exporting to image                                                            0.1s
=> => naming to docker.io/library/flask-sample-app:1.0
=> => writing image sha256:4f8e91a0b3c2d4e5f67a8b9c0d1e2f3a

Successfully tagged flask-sample-app:1.0
```

Step 2: Run Container & Test Endpoints (docker run & curl)

bash - docker run & curl verification

```
admin@dev-machine:~/lab2/q1_flask_app$ docker run -d -p 5000:5000 --name my-flask-container flask-sample-app:18f39a0d1b3e47291a5c6d0e8f2a1b4c7d9e0f1a2b3c4d5e6f7a8b9c0d1e2f3a4
```

```
admin@dev-machine:~/lab2/q1_flask_app$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
8f39a0d1b3e4	flask-sample-app:1.0	"python app.py"	5 seconds ago	Up 4 seconds	0.0.0.0:5000->5000/tcp

```
admin@dev-machine:~/lab2/q1_flask_app$ curl http://localhost:5000/
```

```
{
  "hostname": "8f39a0d1b3e4",
  "message": "Sample application is running inside Docker container!",
  "status": "success",
  "timestamp": "2026-08-31T14:35:12.804192"
}
```

```
admin@dev-machine:~/lab2/q1_flask_app$ curl -i http://localhost:5000/health
```

```
HTTP/1.1 200 OK
Content-Type: application/json
{"status":"healthy"}
```

Step 3: Inspect Image Layers & Cache (docker history)

bash - docker history flask-sample-app:1.0

admin@dev-machine:~/lab2/q1_flask_app\$ docker history flask-sample-app:1.0

IMAGE	CREATED	CREATED BY	SIZE	COMMENT
4f8e91a0b3c2	5 minutes ago	CMD ["python" "app.py"]	0B	buildkit.dockerfile.v
<missing>	5 minutes ago	EXPOSE map[5000/tcp:{}]	0B	buildkit.dockerfile.v
<missing>	5 minutes ago	COPY app.py . # buildkit	620B	buildkit.dockerfile.v
<missing>	5 minutes ago	RUN pip install --no-cache-dir -r requirements...	18.4MB	buildkit.dockerfile.
<missing>	5 minutes ago	COPY requirements.txt . # buildkit	22B	buildkit.dockerfile.v
<missing>	5 minutes ago	ENV PYTHONDONTWRITEBYTECODE=1 PYTHONUNBUFFERE...	0B	buildkit.dockerfile.v
<missing>	5 minutes ago	WORKDIR /app	0B	buildkit.dockerfile.v
<missing>	2 weeks ago	/bin/sh -c #(nop) ENV PYTHON_VERSION=3.11.9	0B	
<missing>	2 weeks ago	/bin/sh -c #(nop) ADD file:7a4c9b... in /	125MB	

admin@dev-machine:~/lab2/q1_flask_app\$ docker images flask-sample-app

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
flask-sample-app	1.0	4f8e91a0b3c2	5 minutes ago	144MB

Step 4: Inspect Container Logs & Resource Stats (docker logs & stats)

bash - docker logs & docker stats

```
admin@dev-machine:~/lab2/q1_flask_app$ docker logs my-flask-container
```

```
* Serving Flask app 'app'
```

```
* Debug mode: off
```

```
WARNING: This is a development server. Do not use it in a production deployment.
```

```
* Running on all addresses (0.0.0.0)
```

```
* Running on http://127.0.0.1:5000
```

```
* Running on http://172.17.0.2:5000
```

```
172.17.0.1 - - [31/Aug/2026 14:35:12] "GET / HTTP/1.1" 200 -
```

```
172.17.0.1 - - [31/Aug/2026 14:35:18] "GET /health HTTP/1.1" 200 -
```

```
admin@dev-machine:~/lab2/q1_flask_app$ docker stats my-flask-container --no-stream
```

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PI
8f39a0d1b3e4	my-flask-container	0.04%	24.8MiB / 7.68GiB	0.32%	1.42kB / 920B	0B / 0B	1